Reinforcement Learning for Stochastic Vaccine Allocation on Contact Networks: Bridging Optimal Control and Individual-Level Dynamics

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I. Statement of the Problem and Objective

The COVID-19 pandemic has demonstrated both the utility and limitations of deterministic models to guide vaccine allocation strategies [1]. The recent work by Arghal et al. introduced an efficient deterministic optimal control framework that allocates vaccines between risk and contact-based population groups using a tractable ODE-based compartmental model [2]. However, this approach inherently assumes statistical homogeneity within groups and neglects stochastic variability and structural differences at the individual level [3].

In practice, these assumptions often break down, particularly in small or structured populations (e.g. nursing homes, classrooms, or essential worker networks), where the law of large numbers does not apply and random fluctuations can significantly impact outcomes [4]. Moreover, deterministic models cannot capture phenomena such as superspreading events, stochastic extinction, or the role of network topology in shaping outbreak dynamics [5], [6]. Studies have shown that in such settings, incorporating individual-level contact heterogeneity and probabilistic transmission dynamics leads to markedly more accurate forecasts and intervention outcomes [7].

This thesis proposes to address these limitations by developing a stochastic epidemic model on a contact graph, where each node represents an individual and each edge encodes a probabilistic contact and infection event. Unlike the compartmental mean-field model, this framework allows for heterogeneity both across and within demographic groups, and captures randomness in contact patterns and disease progression.

*a) **Model Formulation.**:* Let $G = (V, E)$ denote the individual-level contact network, where each node $i \in V$ is an individual, and each edge $(i, j) \in E$ has an associated transmission potential p_{ij}. Each individual i has a group label $g_i \in \{X, Y, Z\}$ corresponding to baseline, high-risk, or high-contact types, determining group-specific parameters (e.g., probability of symptoms s_i, probability of hospitalization π_i, probability of death λ_i).

The epidemic evolves as a continuous-time Markov process (CTMP) over node states $X_i(t) \in \mathcal{S} = \{S, E, P, A, I, L, H, R, V, D\}$, with stochastic transitions governed by exponentially distributed waiting times. For example, the infection rate of a susceptible node i is given by:

$$q^{(i)}_{S\to E} = \sum_{j\in\mathcal{N}(i)} \beta_{ij} \Big(\rho \cdot \mathbf{1}_{\{X_j(t)=P\}} + \mu \cdot \mathbf{1}_{\{X_j(t)=A\}} + \omega \cdot \mathbf{1}_{\{X_j(t)=I\}} \Big)$$

which mirrors the structure of the ODE formulation of Arghal et al. but is applied at the individual level, preserving heterogeneous contact rates and disease trajectories. Other transitions, such as $E \to P$ or $E \to A$, occur with rates τs_i and $\tau(1 - s_i)$, respectively, and hospitalization and mortality outcomes follow group-dependent branching probabilities.

The complete system thus defines a high-dimensional, individual-based CTMP with heterogeneous, graph-structured transition dynamics. When coupled with vaccination policies that control the transition $S \to V$ under constrained resources, this framework gives rise to a stochastic optimal control problem, where the goal is to find a vaccination policy that minimizes an expected loss function, such as cumulative deaths, subject to the stochastic evolution of the system.

*b) **Vaccination and Control.**:* The vaccine policy is encoded as a time-dependent function $u_i(t)$ that specifies the instantaneous vaccination rate for an individual i. The overall vaccination capacity constraint remains:

$$\sum_{i\in V} u_i(t) \cdot \mathbf{1}_{\{X_i(t)=S\}} \le V_0, \quad \forall t.$$

The objective is to minimize cumulative deaths or hospital burden by solving a stochastic optimal control problem:

$$\min_{u(t)} \mathbb{E}\left[\sum_{i\in V} \mathbf{1}_{\{X_i(T)=D\}} \right],$$

subject to the system evolving as a high-dimensional CTMP defined on the network.

*c) **Learning-Based Optimization.**:* Due to the intractability of solving this problem analytically or through dynamic programming, this thesis proposes a reinforcement learning (RL) approach to policy optimization. To improve sample efficiency and inject structural prior knowledge, the optimal deterministic policy derived from the ODE-based model of Arghal et al. will be used as an initial policy to warm-start the RL agent. This allows the learning process to begin with a policy that is provably optimal under mean field assumptions and progressively refine it in the presence of individual-level stochasticity.

The final objective is to combine the structure and scalability of deterministic optimal control with the adaptability and robustness of RL in graph-based epidemic environments, thus allowing the design of practical context-sensitive vaccine allocation strategies under uncertainty.

II. Scope of the Studies

This research aims to develop and evaluate a stochastic vaccine allocation framework based on individual-level epidemic dynamics in contact networks. The scope of the proposed study includes the following components:

- **Thematic Scope:** The study focuses on modeling epidemic spread and optimizing vaccine allocation strategies under stochastic dynamics. It specifically addresses individual-level heterogeneity and probabilistic transmission on contact graphs, going beyond group-based deterministic models.
- **Methodological Scope:** The project integrates epidemic modeling through continuous-time Markov processes, stochastic control theory, and reinforcement learning (RL). RL techniques, particularly policy gradient methods, will be employed to solve the high-dimensional control problem, with warm-start initialization from a known deterministic policy.
- **Model and Data Scope:** Simulated epidemic scenarios will be constructed using synthetic and real-world contact network structures. The model represents individuals as nodes with group-specific disease parameters, and interactions as edges with probabilistic infection rates.
- **Evaluation Scope:** Learned vaccine allocation policies will be evaluated using metrics such as cumulative deaths, infection rates, and peak hospitalizations. Comparisons will be made with deterministic baselines and heuristic allocation rules, alongside sensitivity and ablation analyzes.
- **Exclusions and Assumptions:** The study does not incorporate behavioral factors such as hesitancy in the vaccine, logistical supply chain constraints, or adaptive testing strategies. The vaccination budget is assumed to be known and fixed in advance.

This scope ensures a focused exploration of stochastic control strategies in networked epidemic environments and sets the foundation for future extensions involving real-time adaptivity or partially observable dynamics.

III. Outline of the Proposed Final Document

For an overview of the proposed thesis structure and the corresponding content, refer to Table I.

Section	Content Description
Introduction	Overview of the vaccine allocation problem, motivation for moving beyond deterministic models, and a concise statement of objectives and contributions.
Related Work	Survey of prior efforts in epidemic modeling, optimal control in compartmental settings, and reinforcement learning in healthcare interventions.
Problem Formulation	Definition of the individual-level contact graph, disease states, stochastic dynamics using CTMC, and formalization of the control problem with vaccination constraints.
Methodology	Description of model implementation, simulation environment, reinforcement learning setup, initialization strategies, and optimization constraints.
Experiments and Results	Details of experimental design, evaluation metrics (e.g., mortality, hospitalization), comparative results, ablation analysis, and robustness studies.
Discussion	Critical analysis of findings, interpretability of learned policies, and trade-offs in scalability and computational burden.
Conclusion and Future Work	Summary of key results, implications for epidemic control policy, and directions for extending the model to real-time control, partial observability, and broader disease settings.

TABLE I
PROPOSED STRUCTURE OF FINAL THESIS DOCUMENT

References

[1] D. Olabode, J. Culp, A. Fisher, A. Tower, D. Hull-Nye, and X. Wang, "Deterministic and stochastic models for the epidemic dynamics of covid-19 in wuhan, china." *Mathematical Biosciences and Engineering : MBE*, vol. 18 1, pp. 950–967, 2021.

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